

implemented.¹⁴²⁸ Panel A shows that under the current system with a 30 mil access fee cap for quotations priced \$1.00 or more and a 0.3% access fee cap for transactions less than \$1.00, the exchanges collected an estimated \$3.4 billion in access fees and distributed \$3.1 billion in rebates in 2023, providing an estimated net capture of \$337 million for the exchanges in that time period.¹⁴²⁹ In Table 13 the Commission estimates that the exchanges would have collected \$1,188 million in access fees and distributed \$906 million in rebates in 2023 under the amendment to Rule 610, providing the exchanges a net capture of \$282 million in that time period. Thus, the Commission estimates that total access fees collected would have declined by \$2.23 billion and rebates distributed by \$2.17 billion in 2023.¹⁴³⁰ This amounts to an estimated decline in net capture of \$54.9 million across all exchanges. This decline is conditional upon exchanges maintaining a 2 mil capture rate for stocks trading at a price of \$1.00 or higher. Any estimated changes in total net capture across exchanges is due exclusively to the change in the access fee cap for stocks trading below \$1.00.

Panel B provides estimates of the effect of the amendments on access fees paid and rebates received by liquidity demanders and providers separately. The Commission estimates that, under the amendments, liquidity demanders would have paid \$1.93 billion less¹⁴³¹ in access

¹⁴²⁸ This assumes that exchanges continue the practice of funding rebates through access fees, that trading volumes are unchanged relative to 2023, that the distribution of trading volume across exchanges is unchanged, and that the distribution of trading volume priced below \$1.00 and at or above \$1.00 remains unchanged.

¹⁴²⁹ See Table 6 for additional analysis on current estimates of exchange net capture.

¹⁴³⁰ Balancing out expected rebates paid on make-take, inverted, and flat fee venues, the Commission estimates that liquidity demanders will pay \$1.93 billion per year less in access fees netted across all venues under the Rule and liquidity providers will receive \$1.88 billion per year less in rebates netted across all venues.

¹⁴³¹ The ultimate effect of this change will not result in liquidity demanders saving the full \$1.93 billion in transaction costs, because the effect of reduced rebates will cause the quoted spread to widen, offsetting this reduction in the access fee. See supra section VII.B.3, infra section VII.D.2.c for a discussion of these points.

fees and liquidity providers would have received \$1.88 billion less in rebates in 2023. Thus, the current estimated \$2.6 billion in fee funded rebates in 2023 would have decreased by approximately 70% under the amendments.

Table 13: Estimated Access Fees and Rebates Collected - Current and Adopted 2023^a			
Panel A: Estimated Access Fees Collected and Rebates (in Millions of Dollars)			
	Current	Rule	Difference
Fees Collected	3,414.00	1,188.91	-2,225.21
Rebates Distributed	-3,076.50	-906.16	2,170.30
Exchange Capture	337.66	282.75	-54.90
Panel B: Estimated Fees by Liquidity Type (in Millions of Dollars)			
	Current	Rule	Difference
Liquidity Demander	2,969.12	1,034.61	-1,934.51
Liquidity Provider	-2,631.47	-751.86	1,879.61
Exchange Capture	337.66	282.75	-54.90
^a This table takes trading volumes presented in Table 5 to calculate aggregate fee and rebate estimates under the Rule. Current estimates of fees collected and rebates distributed are taken from Table 6. The analysis presumes that exchanges with fees and rebates currently above 10 mils will decrease fees and rebates to a 10 mil fee and 8 mil rebate (the exceptions being IEX which charges 10 mils to takers and rebates 4 mils to makers, NYSE Chicago which charges both sides 10 mils, and LTSE which does not charge fees). For trading in securities priced less than \$1.00, estimates of fees and rebates presume that all sub \$1.00 fees from Panel B of Table 4 which are over 0.10% are reduced to 0.10%, fees at or below 0.10% remain the same. Computations are made per exchange and then aggregated as shown above.			

Table 14 presents analysis showing an estimated total reduction of approximately \$55 million per year in net capture due to the reduction in the access fee cap and how it might in turn affect the transaction revenues of each of the various exchange families. This estimated decline in transaction revenue comes exclusively from the reduction in the access fee cap for transactions in securities below \$1.00.¹⁴³² This is because, as previously explained, the Commission expects

¹⁴³² This \$54 million estimate is lower than the estimated \$89 million per loss year provided in the Proposing Release. The difference comes because the adopted access fee cap for transactions priced below \$1.00 is

that for transactions priced equal to or greater than \$1.00 the exchanges should be able to maintain their current net capture.¹⁴³³ For transactions priced below \$1.00 most exchanges currently charge the maximum 0.3% but typically offer no rebates.¹⁴³⁴ Because very few exchanges offer rebates on stocks priced below \$1.00, the access fee represents the exchange's net capture. Lowering the access fee from 0.3% to 0.1% on these transactions will represent a decrease in net capture of 66% for many exchanges. This decrease may vary across exchanges. Some exchanges do not charge any fees for trading in sub \$1.00 securities, while others charge a fee to both sides of a sub \$1.00 transactions. Additionally, the exchanges differ in the fraction of sub \$1.00 trading volume that they handle.¹⁴³⁵ Table 14 provides an annual estimate of the effect on exchange transaction revenue of lowering the access fee on exchanges' net capture given realized volumes in 2023 for each exchange group. To the extent that the reduction in the access fee causes more trading at sub-dollar prices, Table 14 overestimates the reduction in exchange transaction revenue.¹⁴³⁶

Table 14: Estimated Effect of Rule on 2023 Exchange Transaction Revenue for Stocks Prices Below \$1.00^a
--

higher than the proposal: 0.10% compared to 0.05% proposed. This reduces the loss on transactions priced below \$1.00. Additionally, as can be seen by comparing Panel B of Table 5 in the Proposing Release and Table 4 herein, multiple exchanges have lowered access fees for transactions below \$1.00 since the proposal making the Rule's difference from the baseline smaller.

¹⁴³³ As discussed in section VII.C.2, the Commission estimates that most exchanges have a net capture of approximately 2 mils on transactions priced greater than \$1.00. For reasons discussed in this section the Commission believes that it is reasonable to assume that exchanges with a current 2 mil net capture would be able to continue to earn a 2 mil net capture.

¹⁴³⁴ See supra Table 4. Most exchanges do not offer rebates for stocks priced less than \$1.00, or if they do the rebates are quite small.

¹⁴³⁵ See supra Table 5.

¹⁴³⁶ The benchmark model in section VII.B.3 implies that a reduction in the access fee will cause the liquidity demand curve to shift, resulting in a higher volume of trades at sub-dollar prices. See also infra note 1462 and surrounding text for a case study on the effect of a rebate instituted by MEMX for sub-dollar trades, which resulted in a higher level of sub-dollar trades; if a reduction in the access fee has a similar effect on equilibrium trading as the institution of the rebate, then the volume of sub-dollar trades will increase and the reduction in exchange revenue will be mitigated.

	Transaction Revenues (\$)	Transaction Revenues (%)
Nasdaq	-\$18,593,052	-20%
NYSE	-\$18,750,074	-19%
Cboe	-\$12,375,769	-16%
MEMX	-\$4,517,207	-21%
IEX	\$0	0%
MIAX	-\$667,838	-7%
LTSE	\$0	0%
Total	-\$54,903,941	-16%
^a The variable Transaction Revenue (\$) provides an annualized estimate of the effect of the amendment to Rule 612 on exchange net capture. For all exchanges, other than LTSE which doesn't charge an access fee and IEX which has an assumed net capture of 6 mils per share traded above \$1.00 (Panel A of Table 4 shows that IEX charges a fee of 10 mils coupled with a rebate of 4 mils), the net capture on transaction priced equal to, or greater than, \$1.00 per share is expected to remain unaffected by the amendments at the assumed 2 mils per share. The 2 mils per share assumption is further discussed in section VII.C.2.c. Thus, the Commission does not expect any decrease in overall exchange transaction revenue per share for shares priced above \$1.00. For transaction volume below \$1.00 per share estimates for the decline in transaction revenue is computed by assuming that under the amendments all exchanges currently charging more than 0.10% for transactions will lower the transaction fee to 0.10%. Exchanges currently charging access fees less than or equal to 0.10% will continue to charge their current rates. The list of current estimated exchange sub \$1.00 pricing comes from Panel B of Table 4. Sub \$1.00 dollar volume estimates for each exchange are from Table 5. The estimated transaction revenue under the amendments is compared to the estimated transaction revenue in the current environment that is estimated using the sub \$1.00 transaction fees/rebates for each exchange presented in Table 4 Panel B and multiplying these fees by volume estimates for each exchange from Table 5. <u>See</u> section VIII.C.2 for Tables 4 and 5. The difference is presented in the Table 14 along with the percent change in transaction revenue from the baseline.		

Lastly, transaction fees in stocks priced less than \$1.00 serve to increase the net cost of accessing liquidity as they do not tend to fund rebates to liquidity providers so there is no incentive that could induce spreads to narrow and on average offset the fee.¹⁴³⁷ Lower transaction costs for these securities may improve liquidity for stocks with prices less than \$1.00.

¹⁴³⁷ See supra section VII.B.3 for additional discussion of how fee-funded rebates are largely off-set by changes in the quoted spread to keep net-costs the same.

However, given the relatively low natural trading interest, the Commission does not expect a significant improvement in the trading environment for these securities.

Given the low net capture rates, the Commission concludes that in most cases, access fees are typically used to fund rebates and not used exclusively to fund execution services. Multiple commenters stated that current access fees and fee caps are not reflective of the current actual costs of providing execution services.¹⁴³⁸ One commenter stated that the cost of processing and matching trades has dropped with technological advances.¹⁴³⁹ Another commenter stated that “the fees charged by exchanges are often far in excess of those necessary to maintain operations of the exchange.”¹⁴⁴⁰ A commenter pointed to significant reductions in spread and commissions since 2005, which have resulted in the 30 mil access fee cap representing a more significant economic factor in trading.¹⁴⁴¹

One commenter stated that a 10 mil access fee cap would represent a “fair pricing model based on the ‘cost plus reasonable return’ methodology” by citing that ATSs charge 10 mil fees while employing similar technologies as exchanges.¹⁴⁴² On this latter point, a commenter stated

¹⁴³⁸ See, e.g., Vanguard Letter at 6, Verret Letter I at 7, and Retirement Coalition Letter at 1.

¹⁴³⁹ See Better Markets Letter I at 15.

¹⁴⁴⁰ See Healthy Markets Letter I at 22.

¹⁴⁴¹ Citing its own previous comment letters one commenter has stated since at least 2014 that a reduction in the access fee cap is warranted given the reduction in trade commissions and narrowing of spreads relative to when the 30 mil access fee cap was first established. See Citigroup Letter at 5.

¹⁴⁴² See Verret Letter I at 5-7. One commenter provided their own review of form ATS-N and specifically looked at minimum and maximum ATS fees; the commenter reported that the maximum ATS fee often exceeds 10 mils by a considerable margin, see Nasdaq Letter III at 3. The Commission does not dispute that maximum ATS fees can exceed 10 mils, but the maximum ATS fee is not an appropriate benchmark for exchange access fees because the maximum ATS fee can be a function of particular services (e.g. block trades or special order types) or of subscriber characteristics (e.g. subscriber order flow might be segmented into specific categories), while an exchange’s access fee schedule applies to all members. Another commenter presented analysis on the subset of ATSs that primarily operate a “continuous book” market and are therefore most closely comparable to exchanges. The commenter’s analysis indicates that seven such ATSs—representing 42% of all ATS volume—charge a maximum of 10 mils. The commenter concluded that the standard rate in the competitive ATS market is 10 mils, while rates substantially above

that a uniform access fee cap of 10 mils would, “have the added benefit of aligning exchange fees with prevailing ATS fees, and creating a more equitable competitive landscape across trading venues.”¹⁴⁴³

In contrast, one commenter stated that the Commission had not established that the “proposed reduced fee caps do, in fact, bear a reasonable relationship to the actual costs to an exchange of a trade.”¹⁴⁴⁴ The same commenter stated that technological costs are not significant determinants of access fee levels, but rather that the fees reflect the magnitude of risk associated with providing liquidity as well as the value to the market that having access to those quotes provides.¹⁴⁴⁵ The commenter further stated that the current access fee cap is not “unreasonably high,” because, among other things, “exchange platform costs to market participants have remained competitive over time.”¹⁴⁴⁶ As the quantitative net capture analysis shows, the access fee cap in the adopted amendments will still permit typical exchange net captures. Thus, for stocks priced greater than \$1.00, lowering the access fee cap is not expected to affect the contribution to revenue and thus to the platform. The following section discusses the effects of a reduction of fees and therefore rebates on the provision of liquidity.

c. Effects on Liquidity and Transaction Costs

The main implications for liquidity from reducing the access fee caps follow the basic principles laid out in sections VII.B.3, as well as the empirical results in section VII.C.2. Most

10 mils are due to specialized services not available on exchanges, see IEX Letter VI at 5; the Commission agrees.

¹⁴⁴³ See BlackRock Letter at 11.

¹⁴⁴⁴ Nasdaq Letter I at 22.

¹⁴⁴⁵ See Nasdaq Letter I at 21; Nasdaq Letter II at 4.

¹⁴⁴⁶ Nasdaq Letter II at 2 and 5.

exchanges charge close to the preexisting access fee cap due, in part, to the disincentive to unilaterally reduce fees and rebates. For the same reasons that exchanges charge close to the preexisting access fee cap, the Commission believes that lowering the access fee cap will lead exchanges to charge similarly close to the new cap. Some exchanges that are currently charging less than the amended access fee cap may continue to do so. Exchanges will most likely not alter their net capture rates, implying that much of the access fee will continue to fund rebates in stocks priced above \$1.00. For reasons discussed further below, the reduction in the access fee cap is likely to leave the cost of accessing liquidity unaffected for some stocks, and to reduce the cost of accessing liquidity for others.

For stocks priced less than \$1.00, the reduction in the access fee cap is also likely to reduce the cost of accessing liquidity. Unlike for stocks priced \$1.00 or more, for stocks priced less than \$1.00 most exchanges charge an access fee without providing a rebate.¹⁴⁴⁷ Since there is no rebate, which would serve to narrow spreads and offset the cost of the access fee, the access fee only serves to increase the cost of taking liquidity in these stocks. Therefore, a reduction in the access fee from 0.3% to 0.1% for stocks priced less than \$1.00 will lower the cost to take liquidity. Some exchanges offer rebates in transactions in stocks priced less than \$1.00.¹⁴⁴⁸ In these instances, under the assumption that these exchanges will uniformly reduce their fees and rebates to maintain the same net capture rate, the reduction in the access fee cap is not expected to affect the cost of taking liquidity.

¹⁴⁴⁷ See supra Table 4.

¹⁴⁴⁸ For example, the Cboe EDGX and MEMX exchanges offer rebates for sub-\$1.00 stocks. See supra Table 4 and surrounding discussion noting that only a few exchanges offer rebates in transactions for stocks priced \$1.00 or less.

Additionally, reducing the access fee cap for stocks priced less than \$1.00 from 0.3% to 0.1% of the quote price will also ensure that the cost of accessing liquidity is similar for stocks with one quote below \$1.00 and another quote equal to or greater than \$1.00. Consider a stock with a best bid quote at \$0.99 and a best ask quote at \$1.00. Under the amendments, the maximum fee to access the bid quote is 9.9 mils and is roughly equal to the 10 mils maximum fee to access the ask quote. Had the Commission lowered the access fee cap for stocks priced \$1.00 or more but left it unchanged at 0.3% for quotes priced less than \$1.00, the cost of accessing the sub-\$1.00 quote would be relatively more expensive than the cost of accessing the \$1.00 or more quote. Here, had the fee cap for quotes priced at or higher than \$1.00 been reduced to 10 mils but the fee cap for sub-\$1.00 trades remained at 0.3%, the maximum allowable fee to access the \$0.99 quote would be 29.7 mils, roughly 3 times greater than that of accessing the ask price. Having a large differential between access fees on opposite sides of an order book would inhibit the ability of markets to reach prices most reflective of the underlying value.¹⁴⁴⁹

Commenters on the proposed access fee cap reduction focused on access fees for stocks priced above \$1.00.¹⁴⁵⁰ Several commenters argued for an alternative in which stocks with a half-penny tick would have an access fee of 15 mils, whereas stocks with a penny tick would have an access fee of 30 mils (hereafter “15 mils/30 mils alternative”).¹⁴⁵¹ As discussed in

¹⁴⁴⁹ Reducing the access fee cap for trades priced at \$1.00 per share or greater to 10 mils without a similar reduction in the fee cap for those priced below \$1.00 could distort markets by introducing an incentive for market participants to exploit differences in fees and rebates for stocks near the \$1.00 threshold. If the maker rebates available under the 0.3% fee cap for sub \$1.00 stocks are greater than those for quotes priced at or greater than \$1.00 then market participants, then market makers may be incentivized to push prices below \$1.00 as they could capture higher rebates by posting at bid and ask at \$0.98 and \$0.99 respectively as opposed to quoting at \$1.00 and \$1.01.

¹⁴⁵⁰ An exception is Cboe Letter IV and Letter II, discussed further at the end of this subsection.

¹⁴⁵¹ See, e.g., NYSE, Schwab, and Citadel Letter at 2, Nasdaq Letter I at 2, MMI Letter at 7, Robinhood Letter at 5, and MEMX Letter at 23-24. See also Nasdaq Letter IV and NYSE Letter I.

sections VII.B.3 and VII.C.2, there is a strong economic tie between the level of the access fee cap and the ability to pay rebates. The discussion among commenters focused on the effect on rebates, with some commenters who favored of the 15 mils/30 mils alternative naming the ability to pay rebates as the primary reason for the higher access fee cap;¹⁴⁵² other commenters specifically were in favor of a ban on rebates.¹⁴⁵³ One of these commenters stated that banning rebates (by requiring exchanges to revert to a pricing model where both sides of a transaction were charged a fee) would fix the problems associated with access fees and rebates, but stated that a second best solution would be to impose a uniform access fee on all exchanges.¹⁴⁵⁴

Commenters in favor of the 15 mils/30 mils alternative expressed the concern that a 10 mils access fee cap would reduce the flexibility to offer rebates. The commenters assert that rebates are necessary to compensate liquidity providers to post displayed (or “lit”) quotes on exchanges.¹⁴⁵⁵ According to the commenters’ logic, a lower access fee cap translates into a lower rebate, which translates into fewer lit quotes. These commenters also state that those quotes that are posted are likely to be wider.¹⁴⁵⁶ Wider and fewer posted quotes, according to these commenters, signify lower market quality.

¹⁴⁵² See, e.g., Nasdaq Letter I at 2.

¹⁴⁵³ See We The Investors Letter I at 3-4 (recommending banning rebates); see also Harris Letter at 4 (recommending reverting to traditional fees, thereby effectively eliminating rebates).

¹⁴⁵⁴ See Harris Letter at 4.

¹⁴⁵⁵ See, e.g., Nasdaq Letter I at 21, Nasdaq Letter II at 5-7, Interactive Brokers Group Letter at 5, Virtu Letter II at 10, Citadel Letter I at 24, WFE Letter at 4, CCMR Letter at 27, and State Street Letter at 4.

¹⁴⁵⁶ See *id.* Some commenters specifically identified the NBBO as a matter of concern (Nasdaq Letter IV at 7; Goldman Letter at 8; Nasdaq letter I at 22, Virtu Letter II at 10.). The NBBO reflects lit quotes at a specific size and thus the arguments regarding the NBBO (with an exception described in more detail below) are the same as those for lit liquidity more generally.

The Commission agrees that lowering the access fee cap is also likely to lower rebates because trading venues use access fees to fund rebates.¹⁴⁵⁷ The Commission also agrees that quoted spreads (spreads that do not reflect rebates or access fees) on lit exchanges are likely to be wider because liquidity providers would be expected to widen spreads to compensate for the lower rebates¹⁴⁵⁸—though the fact that the tick size amendments will lower spreads means that the two amendments combined may in fact lead to lower quoted spreads on some stocks.

The Commission, however, disagrees with the commenters' statements that lower rebates from lower access fees will lower market quality and increase transaction costs. The Commission draws on the economic principles articulated in section VII.B.3.¹⁴⁵⁹ Figure 1 shows how the quoted spreads respond to an equal increase of an access fee and rebate of 30 mils assuming a stock is not tick-constrained. The change contemplated here is a shift of 20 mils because that is the difference between the baseline fee cap of 30 mils and the amended fee cap of 10 mils. When the fees and rebates change together, supply and demand intersect at the same quantity point (thus liquidity offered would be unchanged) but at a different price point, leading to a wider quoted spread. The net spread (the net cost of trading), which takes into account the fees and rebates, would be unchanged.¹⁴⁶⁰ Thus, the Commission disagrees with commenters who argue

¹⁴⁵⁷ See supra section VII.C.2 discussing why trading venues fund rebates with access fees and why rebates are not funded by other revenue sources.

¹⁴⁵⁸ See section VII.B.3 discussing how spreads are expected to widen in response to a reduction in fee-funded maker rebates so to keep the net cost of liquidity constant.

¹⁴⁵⁹ The discussion in that section regarding neutrality of fees and rebates does not depend on the access fee charged per share being equal to the rebate, but rather on fees and rebates being reduced or increased by the same amount. As the Commission does not expect the net capture rate to change, the neutrality result applies.

¹⁴⁶⁰ As explained in section VII.B.3, any change in access fees or rebates may be passed from brokers to customers either directly or indirectly, such as through changes in commissions or changes in the broker's services.